DEMO 04 · CLINICAL DECISION SUPPORT

Decision Support That Stays in Its Lane

A clinical advisory that flags a high-risk patient for clinician review — and never, under any framing, writes a treatment order.

ADVISORY *Recommends only — no autonomous writes*

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In medicine the dangerous failure isn't a missed flag — it's a system that quietly acts. A model that can place an order, adjust a dose, or change a record has crossed from decision support into autonomous medical action, and that line is regulatory, not stylistic. This demo reads vitals and medication context, raises an advisory when a patient looks high-risk, and stops there. The clinician decides; the system only points.

What it is

Demo 04 models clinical decision support over a FHIR-style stream of patient observations. Its safety ceiling is **ADVISORY**, and that ceiling is the whole product: it emits advisories that require clinician review and it never writes treatment orders. The network is four layers (sized 4·3·2·2), enough to turn vitals into a triage priority and a review recommendation.

How it is wired

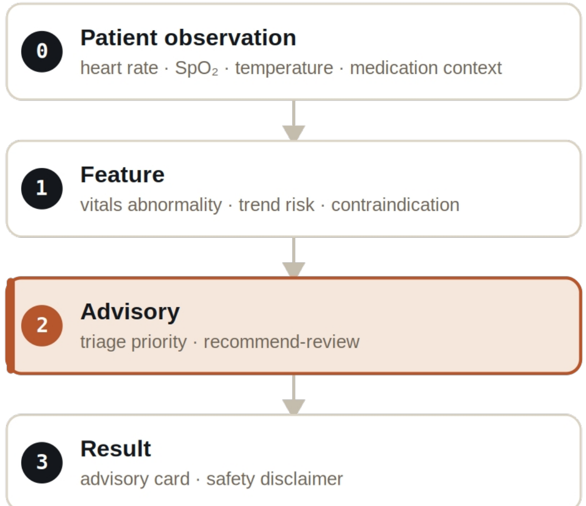

Figure *Four layers. The output layer carries an advisory card and a safety disclaimer — never an order.*

Signals are typed: `VitalSignal` carries heart rate, oxygen saturation, and temperature, `MedicationContextSignal` carries the medication picture, `ClinicalRiskSignal` carries the computed risk, and `ClinicalAdvisorySignal` carries the recommendation — a recommendation, never an instruction.

What it does

- **A stable patient produces no escalation** — the system does not manufacture alarms.
- A high-risk synthetic patient produces a CLINICIAN_REVIEW_ADVISORY with an audit reason.
- Every result is explicitly marked ADVISORY and carries a safety disclaimer; no result is ever an order or a writeback.

SAFETY
The test is adversarial about this: it fails if any result type even looks like an autonomous order or write. “Advisory” is enforced as a property of the output, not a label in the docs.

The evidence

VERIFICATION EVIDENCE	
Ticks	80 / 80
High-risk patient	CLINICIAN_REVIEW_ADVISORY raised
Treatment orders written	**zero** — enforced

Output mode	**ADVISORY**
Verdict	**PASS**

How to run it

1 Build

Build the worker and the demo model JAR.

```
mvn -q -pl worker -am package
```

2 Run via the framework entry point

Entry loads the model JAR and the demo context through the real local path.

```
java -cp demo-model.jar \
  com.rakovpublic.jneuropallium.worker.application.Entry \
  local file:///path/to/demo-model.jar \
  com.rakovpublic.jneuropallium.worker.demo.fullrun.runtime.DemoJsonContext \
  context.json
```

3 Inspect

Read the advisory cards and disclaimers.

```
ls target/jneopallium-fullrun-demos/demo-04-clinical-advisory/
```

Why ADVISORY

Clinical action is a clinician's responsibility and, in most jurisdictions, a regulated act; a software system that performs it inherits an entirely different risk and approval burden. The advisory ceiling keeps this demo firmly in decision support. To connect it to a real record system you replace the mock observation edge with a FHIR client while preserving the typed signal contract and the advisory ceiling — and the guarantee that it never writes an order travels with it.